

ADPS: Asymmetric Distillation Postsegmentation for Image Anomaly Detection

Peng Xing[✉], Hao Tang[✉], Jinhui Tang[✉], *Senior Member, IEEE*, and Zechao Li[✉], *Senior Member, IEEE*

Abstract—Knowledge distillation-based anomaly detection (KDAD) methods rely on the teacher–student paradigm to detect and segment anomalous regions by contrasting the unique features extracted by both networks. However, existing KDAD methods suffer from two main limitations: 1) the student network can effortlessly replicate the teacher network’s representations and 2) the features of the teacher network serve solely as a “reference standard” and are not fully leveraged. Toward this end, we depart from the established paradigm and instead propose an innovative approach called asymmetric distillation postsegmentation (ADPS). Our ADPS employs an asymmetric distillation paradigm that takes distinct forms of the same image as the input of the teacher–student networks, driving the student network to learn discriminating representations for anomalous regions. Meanwhile, a customized Weight Mask Block (WMB) is proposed to generate a coarse anomaly localization mask that transfers the distilled knowledge acquired from the asymmetric paradigm to the teacher network. Equipped with WMB, the proposed postsegmentation module (PSM) can effectively detect and segment abnormal regions with fine structures and clear boundaries. Experimental results demonstrate that the proposed ADPS outperforms the state-of-the-art methods in detecting and segmenting anomalies. Surprisingly, ADPS significantly improves average precision (AP) metric by 9% and 20% on the MVTec anomaly detection (AD) and KolektorSDD2 datasets, respectively.

Index Terms—Anomaly detection (AD), asymmetric distillation, postsegmentation module (PSM), weight mask block (WMB).

I. INTRODUCTION

ANOMALY detection (AD) strives to identify abnormal images and segment the anomalous regions [1], [2], [3], [4], [5], [6]. Its broad applications span from industrial defect detection [4], [7], [8], [9], [10], [11], [12], [13] and medical image diagnosis [14], [15], [16], [17], [18] to the emerging domain of autonomous driving [19], [20]. However, the task of AD is fundamentally different from general classification or segmentation tasks, as anomalies are hard to define and impossible to categorize exhaustively. Therefore, AD models are trained with normal images. Relying only on normal

sample modeling poses a great challenge to detect and segment anomalies.

Substantial advancements in AD have occurred in recent years, with reconstruction-based methods dominating the landscape [10], [21], [22], [23], [24], [25]. The methodology behind these approaches is based on training an autoencoder under a self-supervised paradigm to ensure that out-of-distribution anomalous images are reconstructed with notable errors. The anomalies can then be spotted by assessing the difference between the reconstructed and original image at a high resolution. Innovations such as the self-supervised tasks proposed by Fei et al. [10], focusing on rotation and color prediction, and by Salehi et al. [23], utilizing image patch sorting, have significantly bolstered the efficacy of this approach. Meanwhile, MemAE [21] undermines the generalization ability of the autoencoder by introducing a memory module that enables the recovery of images solely via the stored features. These techniques aspire to inflate the reconstruction error of the anomalous images by increasing the image reconstruction complexity, thereby proving an effective tool to detect out-of-distribution data “unseen” by the model.

Recent studies have shown that knowledge distillation-based anomaly detection (KDAD) methods are promising [4], [12], [13], [26], [27], [28], which enable the student network to learn the consistent representation from the teacher network for normal images. Subsequently, the difference in knowledge representation between the teacher network and the student network is utilized as an AD way, that is, the difference between the features extracted by the teacher–student networks. In the conventional symmetric distillation paradigm, the teacher network’s features are simply used as a “reference standard” and the regions in which the features extracted by the student network do not match them are identified as anomalies, as shown in Fig. 1(a). However, existing advanced methods, such as MKDAD [12] and STPM [29], limit themselves by not delving into the rich knowledge inherent in the pretrained teacher network. Furthermore, they stick to a symmetric distillation paradigm which often leads to over-simulation in the student network, that is, the student model will extract similar representations to the teacher model when confronted with anomalous regions, thereby impeding its ability to generate differentiated representations for anomalous regions. By observing the results of the symmetric distillation models as shown in Fig. 2, we experimentally find that the features generated by the anomaly regions only have slight differences compared to those generated by the normal

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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: xingp_ng@njust.edu.cn; tanghao0918@njust.edu.cn; jinhuitang@njust.edu.cn; zechao.li@njust.edu.cn).

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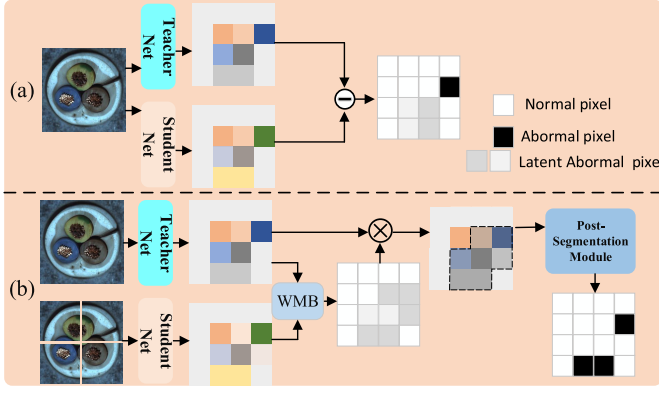


Fig. 1. Schematic of (a) conventional distillation-based symmetric paradigm and (b) proposed asymmetric paradigm.

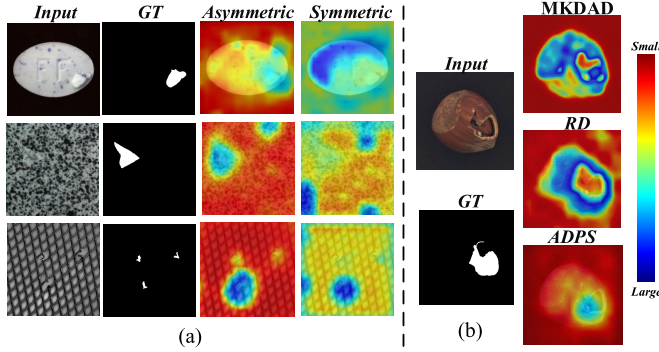


Fig. 2. Visualization of the feature differences extracted by different teacher-student distillation structures. (a) Comparison of using symmetric or asymmetric distillation paradigm in the proposed ADPS. (b) Comparison of ADPS with two representative distillation-based methods: MKDAD [12] and RD [13]. The color of the legend indicates the magnitude of the feature differences in the teacher-student model. Closer to the “Small” side indicates smaller feature differences. On the contrary, closer to the “Large” side indicates large feature differences.

regions, this is particularly problematic when the anomalous regions are similar to the normal regions.

To address the aforementioned issues, we propose a novel asymmetric distillation postsegmentation (ADPS) method, namely ADPS, for image AD, which departs from the common symmetric distillation paradigm by proposing instead to establish a novel asymmetric distillation paradigm. It is inspired by the prior reconstruction-based autoencoder that utilizes asymmetry to augment the complexity of the student network’s alignment with the teacher network. As illustrated in Fig. 1(b), ADPS assembles a teacher-student network with a Weight Mask Block (WMB) and a postsegmentation module (PSM).

First, the proposed asymmetric distillation paradigm is applied by ADPS to the teacher-student networks, allowing the same layer of both networks to process diverse forms of the same data. Specifically, the holistic image is assigned to the teacher network while the student network is given nonoverlapping patches. The insight behind the employed asymmetric inputs is that the student network is encouraged to explore unique expressive abilities different from the teacher network, thus enabling the teacher-student networks to generate more differentiated features when meeting

anomalous samples (see Fig. 2). Meanwhile, smaller local patches can induce the distillation model to focus on identifying minor anomalies within a local region. As a result, the distilled knowledge obtained by asymmetric distillation possesses a more powerful anomaly discrimination ability. Based on the characteristics of the asymmetric distillation paradigm, a customized WMB is proposed to transfer the distilled knowledge to the teacher network, exploiting feature correlations to generate a coarse localization mask. This is expected to improve the discriminability of the teacher model to identify abnormal regions. Contemplating that distilled knowledge may contribute to reducing spurious error and predicting fine anomalies, we develop a novel PSM that implicitly injects the distilled knowledge into the decoding process to further explore normal/abnormal distributions and obtain high-confidence segmentation results with clear boundaries.

Extensive experiments on three benchmark datasets demonstrate the effectiveness of our ADPS, which outperforms the recent state-of-the-art methods on both the challenging MVTec AD dataset [4] and KolektorSDD2 [30], achieving more than 9% and 20% improvement, respectively, in terms of the AP metric for anomaly segmentation. The main contributions of this work are summarized as follows.

- 1) We propose a novel ADPS method, that is, ADPS, which establishes an efficient asymmetric knowledge distillation paradigm to boost the performance of image AD.
- 2) We carefully design the WMB and PSM to incorporate distilled knowledge and improve the representation of discriminative features for the abnormal region.
- 3) Experimental results demonstrate that the proposed ADPS achieves competitive results on three challenging datasets, particularly in solving AD and anomaly segmentation simultaneously.

The remainder of this article is arranged as follows. Section II provides a comprehensive discussion about the literature directly associated with our work. The technical details of the proposed ADPS are presented in Section III. Section IV reports experimental results, ablation studies, and comprehensive analysis. Ultimately, we summarize our findings and suggestions for future research in Section V.

II. RELATED WORK

A. Autoencoder-Based AD

Autoencoder-based methods are a widely used approach for AD, commonly employing different self-supervised pretext tasks for training and identifying anomalies by measuring the pixel-level difference between the original input and the reconstructed output of the model [20], [31]. Examples of such methods include AE-SSIM [31], MemAE [21], MGNAD [22], and DAAD [32], which mainly employ image reconstruction techniques. Other approaches proposed by Li et al. [33] and Zavrtnik et al. [34] utilize image inpainting techniques to complete the content within missing masks, assuming that the model cannot infer anomalous content. However, due to the strong ability of potent pixel-level reconstruction,

these autoencoder-based models sometimes lead to well-reconstructed anomalies, which is referred to as the “identity shortcut” [35]. To prevent the recovery of abnormal images, some studies [21], [22], [32], [36], [37] introduce memory modules that store normal features. Meanwhile, other methods [10], [23] introduce more complex self-supervised tasks, such as coloring and rotation prediction, to detect unseen anomalies that cannot be easily recovered. Despite their strengths, autoencoder-based approaches often struggle with anomaly segmentation due to high reconstruction errors in normal regions [35], [38].

B. Generative Model-Based AD

To overcome the limitations of autoencoder-based methods, research has explored generative model-based approaches, such as GAN [39]), as demonstrated in previous works [40], [41]. In these methods, anomalies are identified if the generated sample significantly differs from the input sample [20], [40]. AnoGAN [40] is one such method that employs noise to generate a target image and then compares the differences between the original and the generated image. However, during the inference phase, this model suffers from a significant drawback involving computationally expensive and time-consuming parameter updating processes. To overcome this issue, EffGAN [42] introduced an encoder that constrains the input noise to its output, successfully alleviating the time-constraint issue. Further refinements in this direction led to GANomaly [41] and f-AnoGAN [14], which improved the generator’s ability to impose constraints on spatial feature reconstruction. In parallel, DefGAN [43] improved the discriminator segment by integrating multiple discriminators. Recently, Hou et al. [32] introduced an additional discriminator to distinguish the generated image from the original one, enhancing the quality of generation. While these developments signify substantial progress, they fail to effectively handle anomalous samples with minuscule anomaly regions.

C. Deep Feature Modeling-Based AD

Another distinctive approach to detecting anomalies involves deep feature modeling-based methods, which assume the existence of a gap between normal and abnormal distributions. These methods begin by constructing a feature space specifically for normal images and then determine whether an image is abnormal by evaluating spatial boundaries [9], [44], [45], [46], [47], [48]. Among these methods, SPADE [48] stands out by deploying a pretrained WideResNet50 [49] to extract patch features. The presence of anomalies is established by determining whether the tested patch contains k adjacent normal patches. PaDiM [9] builds on this method by further refining the process. It utilizes a pretrained model for feature extraction and models normality at each location using a multivariate Gaussian distribution [50], [51]. Further efforts have been made to explore the interrelationships within the data. For instance, Liu et al. [52] proposed a method that models the relationship between a data point and its neighbors. Other approaches have attempted to implicitly model

the distribution. Rudolph et al. [53] and Roth et al. [54], for example, utilize normalizing flow to model the relationship between the original and normal feature distributions. Recent studies [55], [56] similarly seek to model the distribution relationship between patches. Notably, the method presented in [56] introduces visual rotation invariant features and graph networks. Despite these advancements, a significant challenge remains: these methods have not yet delivered effective results in anomaly segmentation.

D. Knowledge Distillation-Based AD

Knowledge distillation-based methods leverage discrepancies in the expressive abilities of the teacher–student networks, thereby enabling them to extract differential features when encountering abnormalities. The pioneering work of U-Std [26] employed a knowledge distillation model for AD. MKDAD [12] was subsequently introduced to address a flaw in U-Std, namely that it only leverages the output of the final layer. Wang et al. [29] proposed the student–teacher feature pyramid matching (STPM), implementing the multiscale feature map approach, which differs from MKDAD by utilizing the discrepancy between multifeature maps to derive localization maps instead of the gradient of the loss function. However, models with symmetric paradigms or similar structures are unable to extract differentiated features. RD [13] employs a pretrained model as an encoder, enabling the student decoder to align with the encoder. However, it merely employs the output of the teacher network as a “reference standard” without further exploiting the strong characteristics of the teacher. Additionally, while RD predominantly excels in detecting anomalies, it falls short in accurately segmenting anomalous regions. To address these limitations, ADPS introduces the asymmetric distillation paradigm. This paradigm enhances symmetric distillation paradigms to concentrate on fine anomalies and simultaneously employs Weighted Mask Block and PSM to amalgamate distilled knowledge with pretrained prior knowledge. This enables a more nuanced exploration of normal and anomalous feature distributions, which is crucial for accurately segmenting fine anomalous regions.

III. PROPOSED APPROACH

As illustrated in Fig. 3, this article proposes a novel method called ADPS, which includes an asymmetric distillation paradigm, a WMB and a PSM specialized for AD. The asymmetric distillation paradigm devises inputs with asymmetry that are utilized to enhance the representation ability gap of the teacher–student networks, thereby generating more discriminative differential features to anomalous regions. The distilled knowledge is further exploited by WMB and injected into the features extracted from the pretrained teacher network. PSM accepts features of the teacher network consisting of prior knowledge and distilled knowledge as input and utilizes the segmentation module to learn the distribution of normal features. The detailed architecture of our ADPS is described as follows.

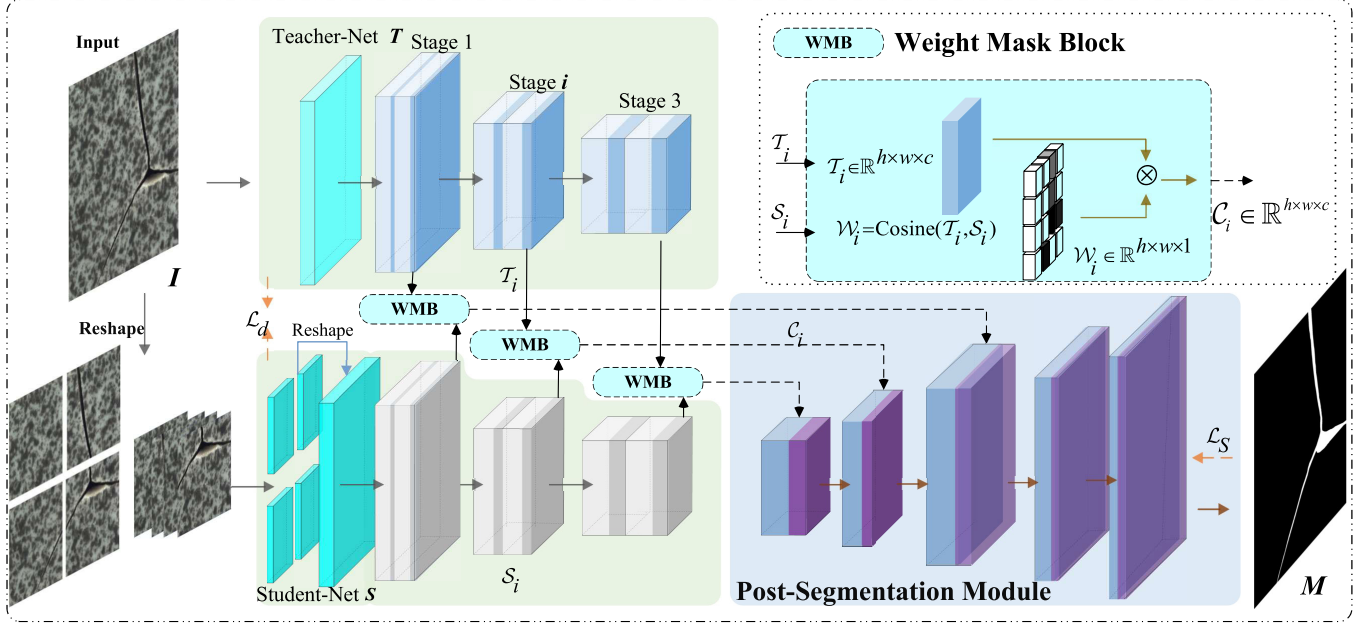


Fig. 3. Overview of the proposed ADPS framework. The asymmetric distillation paradigm uses asymmetric data (e.g., I/I_S and T_i/S'_i) to stimulate S in learning effective discriminative representations. WMB generates a coarse localization mask to transfer distilled knowledge to the teacher model with a powerful representation capability, leading to the feature C_i . Based on WMB, PSM explicitly learns normal and anomaly feature distribution in C_i with a segmentation module to precisely segment finer anomaly regions. In fact, ADPS applied asymmetric inputs to the input layers, Stage 1 and Stage 2 of the student model.

A. Asymmetric Distillation Paradigm

In the field of AD, knowledge distillation has been leveraged to exploit the differential knowledge representation abilities possessed by the teacher and student networks to accurately identify anomalous regions from extracted features. However, a crucial drawback of conventional distillation frameworks, such as the symmetric distillation paradigm depicted in Fig. 2(a), is that the teacher–student network becomes insensitive to disparities in the representation of anomalous regions. This insensitivity stems from the fact that the teacher and student networks tend to learn analogous representations of anomalous regions. Thus, to address these concerns, a feasible solution is to widen the discrepancy in the representational abilities of the teacher–student networks.

The representational abilities of a model are inherently linked to its inputs. Therefore, modifying different inputs for teacher–student networks can obtain different knowledge representation capabilities. We propose an asymmetric distillation paradigm, wherein we construct semantically equivalent, yet structurally different inputs for the teacher–student networks. The disparity in format aims to probe their expressiveness while simultaneously escalating the difficulty of alignment between the student and teacher networks for unfamiliar anomalous regions. As a result, the preservation of the same semantic guarantees that, through training, the student network learns the same features as the teacher network for normal samples. Another key factor is that the local features obtained by the student network are affected only by their own patches and do not stray from anomalous feature vectors in remote parts.

Our ADPS proposes a straightforward yet effective approach for generating distinct but semantically identical data by splitting original images. To implement this approach, a given normal image $I \in \mathbb{R}^{H \times W \times C}$, is split into nonoverlapping local patches, $I_S = \{I_S^1, \dots, I_S^j, \dots, I_S^{k^2}\}$, where $I_S^j \in \mathbb{R}^{(H/k) \times (W/k) \times C}$, and k^2 denotes the number of 2-D patches. In the teacher network, I serves as the direct input, and feature T_i is obtained by a pretrained encoder in stage i . The use of the original image as input in the teacher model assures that T_i does not lose any information about the original image. Meanwhile, in the student network, I_S is utilized as a series of inputs. In fact, this takes k^2 forward processes, and in each forward a patch j of $(H/k) \times (W/k) \times C$ is fed into the student network to extract S'_i . During k^2 times forward, we can obtain I in the feature representation of the student network, that is, $S'_i = \{S'_i^1, \dots, S'_i^j, \dots, S'_i^{k^2}\}$. During the parameter update, the extracted S'_i is subsequently reshaped into S_i based on the original position correspondence, and S_i is calculated as the loss by distillation loss with T_i . The approach, named asymmetric distillation paradigm, could theoretically extend to feature distillation in every intermediate layer by simply splitting features extracted by the intermediate layer.

Since the student network receives a patch of $(H/k) \times (W/k) \times C$ as the input, it will pay more attention to local regions and detect tiny anomalies. Additionally, the asymmetry between local and global receptive fields can exacerbate the complexity of feature alignment in teacher–student networks. The different inputs allow the student network to acquire unique data representation abilities. Therefore, the distilled knowledge of the asymmetric

distillation paradigm presents higher discrimination efficiency between normal and abnormal patterns.

B. Weight Mask Block

Given that prior knowledge in the pretrained teacher network is acquired from an upstream task, such as image classification, it lacks discriminability for normal and abnormal patterns. To further utilize the pretrained a priori knowledge for AD, the proposed WMB transfers distilled knowledge to the teacher network. Formally, “distilled knowledge” refers to the rough localization information of the anomalous region obtained through the asymmetric distillation model. Specifically, features extracted from the teacher and student networks in the asymmetric distillation paradigm are utilized to generate a coarse localization mask (i.e., distilled knowledge, as mentioned above). We then take a weighted approach to explicitly transfer the distilled knowledge to $\mathcal{T}_i$.

An overview of our approach is presented in Fig. 3, where WMB takes $\mathcal{T}_i$ and $\mathcal{S}_i$ as inputs that belong to $\mathbb{R}^{h \times w \times c}$. The correlation of $\mathcal{T}_i$ and $\mathcal{S}_i$ is utilized to generate the coarse localization mask, with the cosine similarity function being used for this purpose. Specifically, the similarity between x and y positions in the feature map is defined as

$$\mathcal{W}_i^{x,y} = \frac{\mathcal{T}_i^{x,y} \cdot \mathcal{S}_i^{x,y}}{\|\mathcal{T}_i^{x,y}\| \times \|\mathcal{S}_i^{x,y}\|} \quad (1)$$

where $\mathcal{T}_i^{x,y} \in \mathbb{R}^{1 \times 1 \times c}$ and $\mathcal{S}_i^{x,y} \in \mathbb{R}^{1 \times 1 \times c}$. The similarity of each position is then evaluated to form the matrix $\mathcal{W}_i = \{\mathcal{W}_i^{x,y} | x \in [1, h], y \in [1, w]\}$. Since the anomalous features in two networks are inclined to be unaligned, $1 - \mathcal{W}_i$ represents the coarse localization of the anomalous regions and serves as an abnormal coarse localization mask. Finally, utilizing explicit weighting, WMB transfers the coarse localization information related to anomalies obtained from distilled knowledge to $\mathcal{T}_i$, with this process being expressed as

$$\mathcal{C}_i = (1 - \mathcal{W}_i) \cdot \mathcal{T}_i \quad (2)$$

where $\mathcal{C}_i$ denotes the outputs generated by WMB. The weighted approach is more suitable for the ADPS than the feature fusion approach, as described in the ablation experiment (Ref. Section IV-D). Therefore, $\mathcal{C}_i$ contains rich knowledge obtained through the pretrained model’s prior knowledge while simultaneously providing the ability to roughly discriminate between abnormal and normal regions based on distilled knowledge.

C. Postsegmentation Module

Traditional KDAD methods often infer the difference obtained from the distillation model as the result of anomaly segmentation. This results in two drawbacks: one is that the features extracted from the pretrained network are only used as a “reference standard” and not explored further for their potential, and another is that the segmentation results obtained yield low resolution with low accuracy. To solve these drawbacks simply and effectively, we leverage the advantages of $\mathcal{C}_i$, which retains both prior and distilled knowledge, and

Algorithm 1 Framework of the Proposed ADPS

Input: Pretrained Teacher model $\mathcal{T}$, student model $\mathcal{S}$, test image set U , Postsegmentation model $\mathcal{PS}$.

Output: Each image’s anomaly map M and its anomaly score Sc .

```

1: for  $I \in U$  do
2:    $I_S \leftarrow Split(I)$   $\triangleright$  Split into nonoverlapping patches
3:    $\mathcal{T}_i \leftarrow \mathcal{T}.forward(I)$ 
4:    $\mathcal{S}_i \leftarrow \mathcal{S}.forward(I_S)$ 
5:   for  $i \in [1, n]$  do  $\triangleright n$  represents the number of layers
6:      $\mathcal{W}_i \leftarrow \mathcal{T}_i, \mathcal{S}_i$  are calculated according to (1)
7:      $\mathcal{C}_i \leftarrow \mathcal{T}_i, \mathcal{W}_i$  are calculated according to (2)
8:   end for
9:    $M \leftarrow \mathcal{PS}.forward([\mathcal{C}_1, \dots, \mathcal{C}_n])$ 
10:   $Sc \leftarrow max(G(M))$   $\triangleright G(\cdot)$  is Gaussian function.
11: end for
```

develop the PSM to identify the normal and abnormal distribution from $\mathcal{C}_i$. PSM seeks to segment the abnormal regions with high accuracy and clear boundaries. Specifically, PSM employs the multiple UpBlock layers of UNet [57] as the backbone. To fully exploit the features extracted from each layer of the pretrained teacher model, we use the weighted multiscale feature map $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_n\}$ as the input. Here, n represents the number of multiscale layers. We present UpBlock i as an example to illustrate the upsampling process.

In UpBlock layer i of PSM, the output $\mathcal{P}_{i-1}$ of UpBlock $i - 1$ is fed as the input

$$\mathcal{P}_i^m = \text{TrConv}_i(\mathcal{P}_{i-1}) \quad (3)$$

where $\text{TrConv}_i(\cdot)$ denotes the $2 \times$ transposed convolution, and $\mathcal{P}_i^m$ is the output of the transposed convolution. Subsequently, $\mathcal{P}_i^m$ and $\mathcal{C}_i$ are concatenated together along the channel dimension

$$\mathcal{P}_i^c = \text{cat}(\mathcal{P}_i^m, \mathcal{C}_i) \quad (4)$$

where $\text{cat}(\cdot)$ denotes the concatenation of two feature maps along the channel dimension. Finally, we obtain the output $\mathcal{P}_i$ of UpBlock i via the convolution layer

$$\mathcal{P}_{i+1} = \sigma(\text{BN}_i^1(\text{Conv}_i^1(\sigma(\text{BN}_i^2(\text{Conv}_i^2(\mathcal{P}_i^c)))))) \quad (5)$$

where $\text{Conv}_i^1(\cdot)$ and $\text{Conv}_i^2(\cdot)$ indicates two 3×3 convolution layers with stride 1 in UpBlock i layer, $\text{BN}_i^1(\cdot)$ and $\text{BN}_i^2(\cdot)$ denotes two batch normalization layers, $\sigma(\cdot)$ denotes the ReLU activation function. Finally, the output $\mathcal{P}_n$ is convolved by 3×3 convolution layer to obtain the segmentation mask M with the size of $H \times W$

$$M = \text{Softmax}(\text{Conv}_f(\mathcal{P}_n)) \quad (6)$$

where $\text{Softmax}(\cdot)$ denotes the Softmax function and $\text{Conv}_f(\cdot)$ represents 3×3 convolution layer.

In Algorithm 1, we show each step of the ADPS framework. With ADPS, we can infer the anomaly score and anomaly segmentation map of the image to be tested.

D. Loss Function

To train ADPS, this work follows the DRAEM [58] to generate simulated anomaly samples. Additionally, distillation loss and focal loss [59] are introduced as part of the loss function. The distillation loss, defined as follows, is used in the model training

$$\mathcal{L}_d = \sum_{i=1}^n (1 - Y) \cdot \left(1 - \frac{T_i \cdot S_i}{\|T_i\| \times \|S_i\|}\right) + Y \cdot \frac{T_i \cdot S_i}{\|T_i\| \times \|S_i\|} \quad (7)$$

where Y denotes the ground-truth (GT) label of training samples such that $Y = \{y_{u,v} \in \{0, 1\} | u \in [1, h], v \in [1, w]\}$. Indeed, $\mathcal{L}_d$ essentially expects the teacher-student model to output features close to each other when fed normal samples and away from each other when fed forged anomaly samples. Furthermore, the proposed approach utilizes the focal loss $\mathcal{L}_s$ [59] defined as follows:

$$\mathcal{L}_s = \begin{cases} -(1 - p_{i,j})^\tau \log(p_{i,j}), & y_{i,j} = 1 \\ -p_{i,j}^\tau \log(1 - p_{i,j}), & y_{i,j} = 0 \end{cases} \quad (8)$$

where $p_{i,j}$ denotes the probability of the abnormality in the coordinate (i, j) of M . To incorporate the distillation loss $\mathcal{L}_d$ and segmentation loss $\mathcal{L}_s$ in the training process of ADPS, the overall loss $\mathcal{L}$ is defines as follows:

$$\mathcal{L} = \mathcal{L}_d + \lambda \mathcal{L}_s \quad (9)$$

where λ is set to control the importance of the two losses.

IV. EXPERIMENTS

A. Experimental Settings

1) *Datasets*: To demonstrate the effectiveness of our proposed ADPS, we conducted experiments on three datasets: MVTec AD [4], KolektorSDD [64], and KolektorSDD2 [30].

- 1) The MVTec AD [4] dataset consists of 15 classes of high-resolution images derived from actual industrial scenarios, subdivided into five categories of texture images (referred to as “Texture”) and ten categories of object images (referred to as “Object”). The training dataset contains only normal images, while the testing dataset includes a variety of anomaly patterns, such as scratches, defects, and so on. This dataset presents a considerable challenge due to the subtle anomalies and high-resolution nature.
- 2) The KolektorSDD [64] dataset consists of 399 images, each with dimensions of 500×1250 . These images were acquired under controlled conditions in a real-world industrial setting.
- 3) The KolektorSDD2 [30] is a comprehensive surface defect detection dataset containing over 3000 images, with each having a size of approximately 230×630 . This dataset exposes a range of defect types commonly found in the industry, including scratches, dots, and surface defects.

2) *Implementation Details*: The input image resolution is fixed at 256×256 , while k is set to 8. In the training phase, the learning rate is set to 0.0001 and the batch size is set to 32. The network is trained for 300 epochs by the Adam optimizer [65], with a learning rate decay at epochs 240 and 270 with a decay rate of 0.2. We set the value of λ to 1.

3) *Baseline Approaches*: We compare our ADPS against several baselines, including: U-Std [26], MKDAD [12], MF [61], RD [13], DAAD+ [32], RIAD [34], Cutpaste [62], DRAEM [58], DRAEM + SSPCAB [66], AnoSeg [63], SGSF [67], SPADE [48], Patchcore [54], PaDim [9], MAD [60], and Semi-orthogonal [68]. Among these, SPADE, PaDim, Semi-orthogonal, Patchcore, and RD utilize WideResNet50 [49] as the backbone, while MF introduces a Transformer-based structure. DRAEM, DRAEM + SSPCAB, Cutpaste, and our ADPS introduce simulated anomalous samples.

4) *Evaluation Metrics*: As per previous studies [9], [13], [69], commonly used evaluation metrics for anomaly classification tasks is the area under the receiver-operating characteristic (AUROC), and for anomaly segmentation tasks are per-region-overlap (PRO) and AUROC. However, these metrics inadequately reflect the actual segmentation performance when dealing with small anomalous regions due to the overwhelming number of normal pixels. Toward this end, Tao et al. [69] introduce a more convincing metric, average precision (AP), to reflect the performance of the proposed method for pixel-level segmentation. In this work, the performance of anomaly classification is evaluated using $AUROC_{cla}$, while anomaly segmentation performance is measured using $AUROC_{seg}$, PRO_{seg} , and AP_{seg} collectively. Notable, on account of the significantly lower number of abnormal pixels than normal ones, AP_{seg} better indicates the segmentation accuracy.

B. Comparison With State-of-the-Art Methods

1) *MVTec AD*: Table I illustrates the anomaly classification results obtained on the MVTec AD dataset. ADPS achieves very competitive anomaly classification results, with 100% classification accuracy achieved across four categories. Significantly, ADPS shows notable performance improvement in several categories, outperforming the methods employing WideResNet-50 with improved accuracy by approximately 3%–10%. These results demonstrate that ADPS makes full use of the pretrained model’s knowledge to achieve efficient anomaly classification. In comparison to Cutpaste [62], which further introduces simulated anomalous samples, ADPS obtains a superior performance gain of 2.7%. Thus, the strategy of leveraging asymmetric distillation to achieve coarse localization and fine segmentation of anomalies proves more effective than training the classifier directly. Overall, the experimental results highlight that ADPS, integrating distilled and prior knowledge, provides a more accurate normal image distribution and delivers powerful anomaly classification capability.

In addition to anomaly classification, we demonstrate the anomaly segmentation performance of ADPS and compare it

TABLE I
ANOMALY CLASSIFICATION RESULTS IN TERMS OF $AUROC_{CLA}$ (%) ON THE MVTEC AD DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Methods	Textures						Objects											MEAN
	Carpet	Grid	Leather	Tile	Wood	mean	Bottle	Cable	Capsule	Hazelnut	Metal nut	Pill	Screw	Toothbrush	Transistor	Zipper	mean	
U-Std [26]	91.6	81.0	88.2	99.1	97.7	91.5	99.0	86.2	86.1	93.1	82	87.9	54.9	95.3	81.8	91.9	85.8	87.7
SPADE [48]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	85.5
MKDAD [12]	79.3	78	95.1	91.6	94.3	87.7	99.4	89.2	80.5	98.4	73.6	82.7	83.3	92.2	85.6	93.2	87.8	87.7
DAAD+ [32]	86.6	95.7	86.2	88.2	98.2	91.0	97.6	84.4	76.7	92.1	75.8	90	98.7	99.2	87.6	85.9	88.8	89.5
RIAD [34]	84.2	99.6	100.0	98.7	93	95.1	99.9	81.9	88.4	83.3	88.5	83.8	84.5	100.0	90.9	98.1	89.9	91.7
MAD [60]	95.5	92.9	100.0	97.4	97.6	96.7	100.0	94.0	92.3	98.7	93.1	83.4	81.2	95.8	95.9	97.9	93.2	94.4
MF [61]	94.0	85.9	99.2	99.0	99.2	95.5	99.1	97.1	87.5	99.4	96.2	90.1	97.5	100.0	94.4	98.6	96.0	95.8
Cutpaste [62]	93.1	99.9	100.0	93.4	98.6	97.0	98.3	80.6	96.2	97.3	99.3	92.4	86.3	98.3	95.5	99.4	94.4	95.2
PaDim [9]	99.8	96.7	100.0	98.1	99.2	98.8	99.9	92.7	91.3	92.0	98.7	93.3	85.8	96.1	97.4	90.3	93.8	95.5
AnoSeg [63]	96.0	99.0	99.0	98.0	99.0	98.2	98.0	98.0	84.0	98.0	95.0	87.0	97.0	99.0	96.0	99.0	95.1	96.0
ADPS	97.4	100.0	100.0	99.8	97.7	99.0	100.0	93.7	96.0	99.6	99.7	95.3	89.5	95.3	97.6	100.0	96.7	97.4

TABLE II
ANOMALY SEGMENTATION RESULTS IN TERMS OF $AUROC_{SEG}$, PRO_{SEG} , AND AP_{SEG} (%) ON THE MVTEC AD DATASET. THE PROPOSED ADPS OUTPERFORMS RECENT METHODS BY NEARLY 10% IN TERMS OF AP_{SEG} . THE BEST RESULTS ARE MARKED IN BOLD.
PADIM* DENOTES ANOTHER RESULT FROM [9] WITH RESNET-18 AS THE BACKBONE

Metric	Methods	Textures						Objects											MEAN
		Carpet	Grid	Leather	Wood	Tile	mean	Bottle	Cable	Capsule	Hazelnut	Metal nut	Pill	Screw	Toothbrush	Transistor	Zipper	mean	
$AUROC_{seg}$	U-Std [26]	93.5	89.9	97.8	92.1	92.5	93.2	97.8	91.9	96.8	98.2	97.2	96.5	97.4	97.9	73.7	95.6	94.3	93.9
	RIAD [34]	96.3	98.8	99.4	85.8	89.1	93.9	98.4	84.2	92.8	96.1	92.5	95.7	98.8	98.9	87.7	97.8	94.3	94.2
	Cutpaste [62]	98.3	97.5	99.5	95.5	90.5	96.3	97.6	90	97.4	97.3	93.1	95.7	96.7	98.1	93	99.3	95.8	96.0
	SPADE [48]	97.5	93.7	97.6	88.5	87.4	92.9	98.4	97.2	99.0	99.1	98.1	96.5	98.9	97.9	94.1	96.5	97.6	96.5
	PaDim* [9]	98.9	94.9	99.1	91.2	93.6	95.6	98.1	95.8	98.3	97.7	96.7	94.7	97.4	98.7	97.2	98.2	97.3	96.7
	PaDim [9]	99.0	98.6	99	94.1	94.1	97.0	98.2	96.7	98.6	98.1	97.3	95.7	94.4	98.8	97.6	98.4	97.4	97.4
	DRAEM [58]	95.5	99.7	98.6	96.4	99.2	97.9	99.1	94.7	94.3	99.7	99.5	97.6	97.6	98.1	90.9	98.8	97.0	97.3
	SGSF [67]	97.4	99.2	98.9	97.3	99.7	98.5	96.3	93.3	97.4	99.5	99.5	99.5	97.2	99.6	84.7	98.2	96.5	97.2
	MKDAD [12]	95.6	91.8	98.1	82.8	84.8	90.6	96.3	82.4	95.9	94.6	86.4	89.6	96.0	96.1	76.5	93.9	90.8	90.7
	Patchcore [54]	98.5	97.4	99.2	96.0	96.8	97.6	98.6	98.6	99.0	98.8	98.6	98.3	98.4	98.7	96.4	98.6	98.4	98.1
	RD [13]	98.9	99.3	99.4	95.6	95.3	97.7	98.7	97.4	98.7	98.9	97.3	98.2	99.3	99.1	92.5	98.2	97.8	97.8
PRO_{seg}	ADPS	99.5	99.2	99.9	99.3	99.6	99.5	99.5	94.6	94.1	99.6	97.5	99.3	98.7	99.1	92.2	99.6	97.4	98.1
	U-Std [26]	87.9	95.2	94.5	94.6	91.12	92.7	93.1	81.8	96.8	96.5	94.5	96.1	94.2	93.3	66.6	95.1	90.8	91.4
	MF [61]	87.8	86.5	95.9	88.1	84.8	88.6	88.8	93.7	87.9	88.6	86.9	93.0	95.4	87.7	92.6	93.6	90.8	90.1
	SPADE [48]	94.7	86.7	97.2	75.9	75.9	87.4	95.5	90.9	93.7	95.4	94.4	94.6	96.0	93.5	87.4	92.6	93.4	91.7
	PaDim* [9]	96.0	90.9	97.9	81.6	90.3	91.3	93.9	86.2	91.9	91.4	81.9	90.6	91.3	92.3	80.2	94.7	89.4	90.1
	PaDim [9]	96.2	94.6	97.8	86.0	91.1	93.2	94.8	88.8	93.5	92.6	85.6	92.7	94.4	93.1	84.5	95.9	91.6	92.1
	RD [13]	97.0	97.6	99.1	90.6	90.9	95.0	96.6	91.0	95.8	95.5	92.3	96.4	98.2	94.5	78.0	95.4	93.4	93.9
	Patchcore [54]	95.9	89.1	97.4	93.9	85.7	92.4	95.6	98	94.6	92.4	93.5	94.7	92.8	93.9	96.7	96.3	94.9	94.0
	ADPS	97.5	97.2	99.3	96.6	97.9	97.7	97.8	82.9	93.4	95.2	91.5	96.0	94.3	93.4	85.4	98.3	92.8	94.4
	U-Std [26]	52.2	10.1	40.9	53.3	65.3	44.4	74.2	48.2	25.9	57.8	83.5	62.0	7.8	37.7	27.1	36.1	46.0	45.5
AP_{seg}	RIAD [34]	61.4	36.4	49.1	38.2	52.6	47.5	76.4	24.4	38.2	33.8	64.3	51.6	43.9	50.6	39.2	63.4	48.6	48.2
	PaDim [9]	60.7	35.7	53.5	46.3	52.4	49.7	77.3	45.4	46.7	61.1	77.4	61.2	21.7	54.7	72.0	58.2	57.6	55.0
	Patchcore [54]	66.7	41	51	59.3	52.3	54.1	80.1	70	48.1	61.5	88.8	78.7	41.4	51.6	63.2	64	64.7	61.2
	DRAEM [58]	53.5	65.7	75.3	77.7	92.3	72.9	86.5	52.4	49.4	92.9	96.3	48.5	58.2	44.7	50.7	81.5	66.1	68.4
	DRAEM +SSPCAB [66]	59.4	61.1	76.0	95.0	77.1	73.7	87.9	57.2	50.2	92.6	98.1	52.4	72.0	51.0	48.0	77.1	68.7	69.9
	RD [13]	64.1	47.6	52.4	50.4	53.6	53.6	79.4	59.2	45.8	64.5	80.9	80.0	54.8	54.5	55.7	60.6	63.5	60.2
	ADPS	83.9	62.3	81.6	89.4	96.3	82.7	92.9	65.4	62.5	89.4	82.9	88.7	59.1	67.5	53.1	85.4	74.7	77.4

to current state-of-the-art methods. The evaluation is based on three key metrics on the MVTEC AD dataset, with the results presented in Table II. Our approach, ADPS, outperforms knowledge distillation-based methods such as U-Std and MKDAD by 5%–10% on $AUROC_{seg}$ and 3% on PRO_{seg} . It also exceeds U-Std by 32% on AP_{seg} , demonstrating its excellent anomaly segmentation capability.

Compared to recent distillation methods, the superior performance of ADPS is attributed to its asymmetric distillation paradigm, which shows effectiveness in handling anomalous images. The representations extracted from the student network of input local images are harder to match with those of the teacher network of input global images. This leads to better detection of intricate anomalous regions,

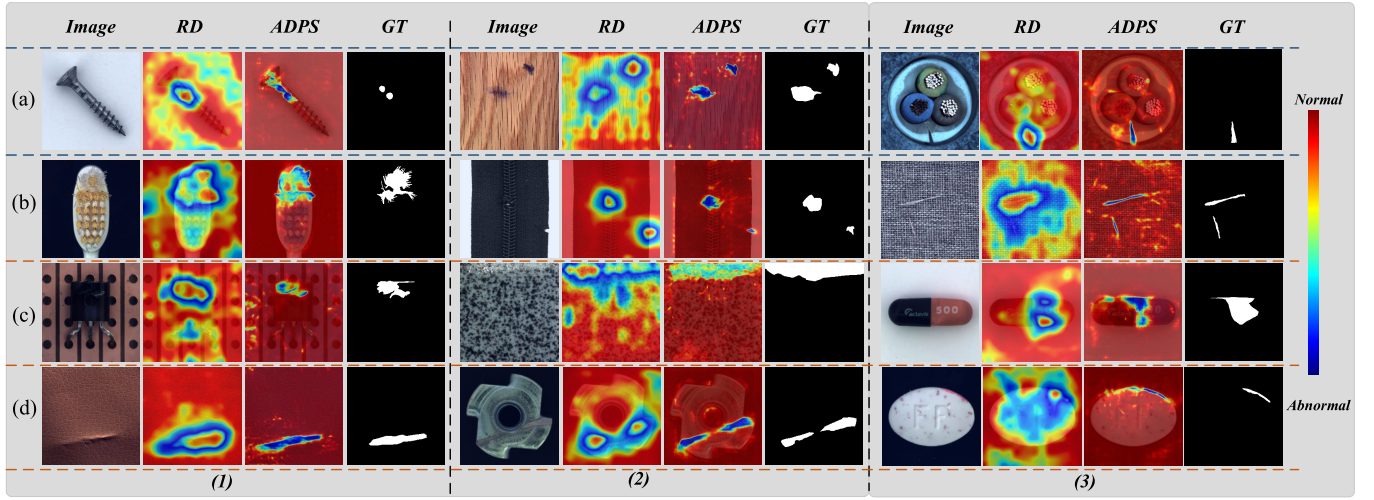


Fig. 4. Anomaly segmentation results on the MVTec AD dataset. Compared with RD [13], ADPS achieves high-confidence segmentation results in both anomalous and normal regions with different categories. (a)–(d) represent the horizontal coordinates and (1)–(3) represent the vertical coordinates in Fig. 4. In total, 12 examples are listed to compare the proposed ADPS with RD.

TABLE III

PERFORMANCE COMPARISON WITH THE LATEST UNSUPERVISED ANOMALY DETECTION METHODS AND SUPERVISED ANOMALY DETECTION METHODS ON THE MVTec AD DATASET

Methods	Ref & Year	Type	$AUROC_{cla}$	$AUROC_{seg} / PRO_{seg} / AP_{seg}$
DeSTSeg [70]	CVPR2023	unsupervised	98.0	97.6 / 95.8 / 75.8
RN [7]	ICCV2023	unsupervised	99.6	98.1 / 94.5 / 59.6
BGAD [71]	CVPR2023	unsupervised	97.4	98.2 / 95.5 / -
ADPS (Ours)	-	unsupervised	97.4	98.1 / 94.4 / 77.4
DRA [72]	CVPR2022	supervised	96.1	85.3 / 73.3 / 26.0
PRN [73]	CVPR2023	supervised	99.4	99.0 / 96.1 / 78.6

TABLE IV

ANOMALY DETECTION RESULTS FOR THE KOLEKTORSDD AND KOLEKTORSDD2 DATASETS

Method	KolektorSDD	KolektorSDD2		
	$AUROC_{seg}$	$AUROC_{cla}$	$AUROC_{seg}$	AP_{seg}
Semi-orthogonal [68]	96.0	-	98.1	-
PaDim [9]	94.5	-	95.6	-
U-Std [26]	89.6	-	95.0	-
DRAEM [58]	-	93.1	93.4	50.5
SGSF [67]	-	93.5	91.5	51.6
RD [13]	-	94.8	98.2	47.7
ADPS	96.3	95.6	99.2	72.5

resulting in a significant improvement in segmentation precision.

Meanwhile, ADPS significantly outperforms Patchcore [54] in its ability to localize anomalies at the pixel level, for example, the AP_{seg} metric outperforms it by 16%. This is due to the powerful anomaly segmentation capability of the ADPS PSB after obtaining distillation knowledge. Second, Patchcore takes more time in inference because it computes anomaly scores based on the similarity measure between an image's patch features and each stored memory bank item. ADPS also surpasses DRAEM, DRAEM + SSPCAB, and Cutpaste methods that rely on forged anomalous samples, even though DRAEM (including combined SSPCAB) have larger parameters of networks, achieving an impressive AP_{seg} of 77.4%. In particular, DRAEM similarly uses a segmentation module, while ADPS significantly outperforms its anomaly segmentation capability. The competitive results of ADPS are attributed to its integration of two fundamental components: harnessing the feature representation capabilities inherent in the teacher network and using distilled knowledge to obtain a coarse localization of anomalous regions followed by precise anomaly segmentation.

Fig. 4 shows the visualization image of anomaly segmentation on the MVTec AD dataset. *a1*–*a3* demonstrate the anomalous region segmentation capability of RD and

ADPS. ADPS utilizes the segmentation module to improve its segmentation accuracy by exploiting coarse localization features from the distilled knowledge. Conversely, RD approximates the location of anomalous regions but is accompanied by significant noise. *c1*, *d2*, and *d3* demonstrate the high resemblance between anomalous and normal regions, which poses challenges to the efficient segmentation of current AD methods. However, the exceptional ability of ADPS in anomaly segmentation stands out.

In addition, we compare the proposed ADPS with recent state-of-the-art unsupervised and fully supervised AD methods. The comparison results are demonstrated in Table III. Compared with the state-of-the-art unsupervised methods, the proposed ADPS is more challenging to localize the anomalies, especially in the metric AP_{seg} , which reaches the highest 77.4%. Compared to the AD method RN [7] based on the knowledge distillation method, ADPS exceeds 17% on the AP_{seg} metric. Compared to the fully supervised method DRA [72], the proposed ADPS leads in all aspects. Undeniably, ADPS still has room for improvement for the state-of-the-art fully supervised method PRN [73]. However, the anomaly segmentation aspect also achieves a performance that wants to be close. The above experimental results demonstrate the advancement of ADPS.

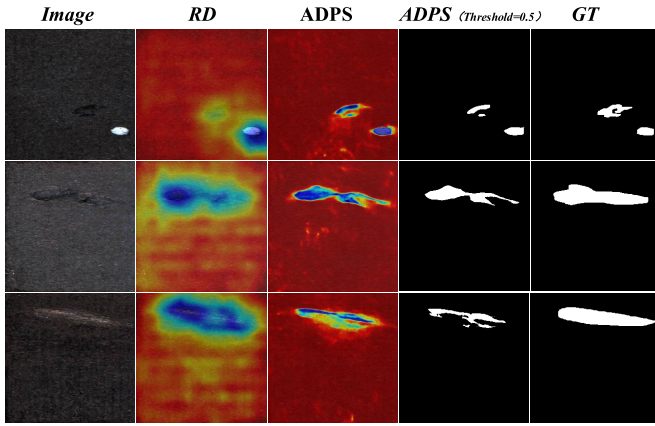
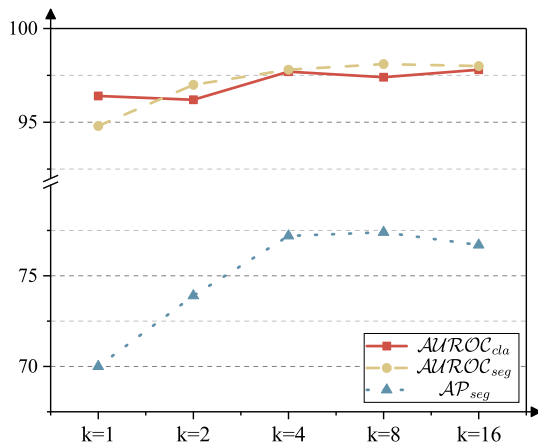
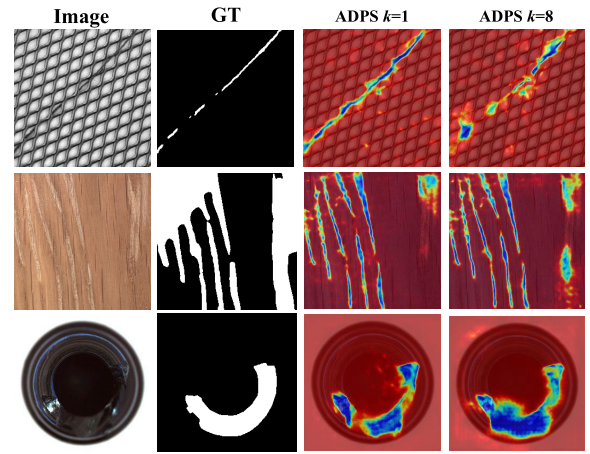


Fig. 5. Anomaly segmentation results on the KolektorSDD2 dataset.

Fig. 6. Performance of asymmetric distillation paradigm ($k = 2, 4, 8, 16$) and symmetric distillation paradigm ($k = 1$).

2) *KolektorSDD&KolektorSDD2*: To further demonstrate the effectiveness of ADPS, we conduct anomaly classification and anomaly segmentation experiments on two additional benchmark datasets (i.e., KolektorSDD and KolektorSDD2). The results, presented in Table IV, show that ADPS achieves state-of-the-art performance on both datasets. Notably, for the KolektorSDD2 dataset, ADPS achieves an AP_{seg} of 72.5%, which is a substantial improvement of 20% and 21% over the performance of DRAEM and SGSF, respectively. It is worth mentioning that both DRAEM and SGSF employ forged anomaly samples and segmentation frameworks. This significant performance advantage can be attributed to the thorough exploration of the potential of pretrained models by ADPS for AD tasks. To achieve accurate anomaly segmentation, distilled knowledge, and prior knowledge are utilized to generate coarse localization information, which is further utilized by the segmentation module. The results of anomaly segmentation are shown in Fig. 5. Exemplary segmentation of anomaly contours and normal regions, along with high-confidence segmentation results, is achieved by ADPS, outperforming RD in this regard. Furthermore, by uniformly calibrating the segmentation threshold at 0.5 across all categories, ADPS obtains precise anomaly segmentation maps.

Fig. 7. Visualization images using symmetric distillation ($k = 1$) and asymmetric distillation methods ($k = 8$).

C. Effectiveness of Asymmetric Distillation Paradigm

1) *Asymmetric Inputs*: We first conduct experiments on the MVTec AD dataset to validate the influence of various inputs on the asymmetric distillation paradigm. When $k = 1$, ADPS adopts the symmetric distillation paradigm. In the cases where k belongs to $\{2, 4, 8, 16\}$, ADPS utilizes the asymmetric distillation paradigm by dividing the input into $k \times k$ patches at the input layer, respectively. The experimental results are illustrated in Fig. 6. The supremacy of the asymmetric distillation paradigm is visibly established, particularly in terms of AP_{seg} . Moreover, the optimal performance is achieved when $k = 4$ or 8 . The division into patches facilitates the focused attention of the student network on specific regions, enabling ADPS to recognize anomalous pixels that closely resemble normal ones. However, excessive division (when $k = 16$) can cause a loss of crucial original structural information leading to a decrease in performance. We visualize the anomaly segmentation results for the symmetric distillation paradigm and the asymmetric distillation paradigm shown in Fig. 7. It is observed that the student network with an asymmetric input can better capture the anomalous features of tiny regions with detailed boundaries. In scenarios like the ‘Grid’ category, the sequential learning of local images allows the student network to concentrate more intensely on local details, thereby facilitating a more accurate localization map for “foreign objects.”

2) *Asymmetric Layers*: In addition to introducing the asymmetric inputs, ADPS introduces the asymmetric distillation paradigm at various feature levels in different stages. Taking WideResNet-50 as the backbone of the teacher network as an example, we introduce the asymmetric distillation strategy in Stage 1 and Stage 2, building upon the asymmetric input layer. The AD results are shown in Table V. The introduction of asymmetric distillation at the shallow layer (stage 1) improves the AD performance for texture images, although its effect on the “Object” type is not substantial. However, the performance drops significantly upon further adoption at the deep layers (stage 2). This may occur due to the low resolution of segmented features in the deeper layers, which disrupts

TABLE V

RESULTS OF INTRODUCING ASYMMETRIC KNOWLEDGE PARADIGM FOR DIFFERENT STAGES WITH $k = 4$. “STAGE i ” INDICATES THAT AT STAGE i

Input Layer	Stage 1	Stage 2	$AUROC_{cla}$			$AUROC_{seg}$			PRO_{seg}			AP_{seg}		
			Texture	Object	Mean	Texture	Object	Mean	Texture	Object	Mean	Texture	Object	Mean
✓	✗	✗	98.2	97.4	97.7	99.5	97.0	97.8	97.8	90.6	93.0	83.1	74.2	77.2
✓	✓	✗	99.3	97.4	98.0	99.5	96.7	97.6	98.0	91.0	93.3	83.7	73.5	76.9
✓	✓	✓	99.1	97.3	97.8	99.4	96.4	97.4	98.0	89.9	92.6	83.1	72.3	75.9

the original nearest-neighbor relationships within the image. Consequently, the destruction of structural information impairs the effectiveness of distillation in providing coarse localization information, thereby influencing the detection capability of ADPS.

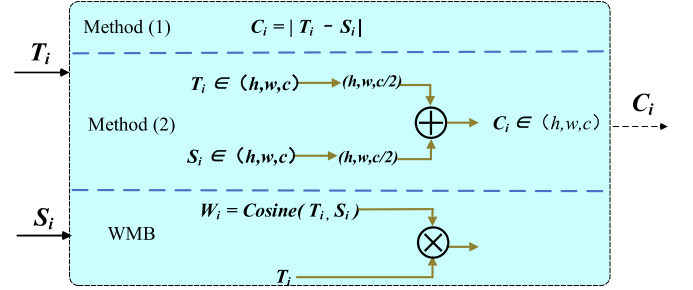
D. Effect of Weight Mask Block on ADPS

We investigate three ways for combining distilled knowledge and pretrained knowledge to obtain C_i , as illustrated in Fig. 8.

- 1) *Method (1)*: This method involves utilizing the differences in features between the teacher network and the student network as C_i , which is then directly input into PSM.
- 2) *Method (2)*: The features extracted from the student network and the teacher network are combined directly, where a 3×3 convolution is employed to achieve dimensional reduction and fusion for obtaining C_i .
- 3) *WMB*: The distilled knowledge $\mathcal{W}_i$ is leveraged in WMB to re-weight the original pretrained representations, resulting in C_i .

The experimental results corresponding to the three ways are shown in Table VI. Comparing Method (1) and Method (2) reveals that the AD capability of Method (1) significantly surpasses that of Method (2). This observation demonstrates that explicitly incorporating coarse localization information outperforms direct feature fusion. Accordingly, we designed a more effective weighted method called WMB. WMB integrates not only the coarse localization information $\mathcal{W}_i$, but also the multiscale contextual information extracted from the pretrained teacher network T_i . WMB exhibits superior performance compared to Method (1), establishing it as the suitable module within the ADPS framework. Despite the simplicity of this weighted approach, it provides a solution for fusing distilled knowledge and pretrained prior knowledge for high-performance AD. Its advantage lies in ingeniously integrating distilled knowledge and pretrained knowledge, thereby enabling PSM to derive fused features optimized for AD.

In addition, we investigated the effect of different distillation losses on ADPS performance in WMB. We compare the cosine similarity (Cosine) loss with the mean squared error (MSE) loss. In Table VII, we show four setups, where the column ‘ $\mathcal{W}_i$ ’ contains the two ways to compute $\mathcal{W}_i$. The experimental results are shown in Table VII. When using both MSE and Cosine similarity losses, the AD performance is reduced. Employing the Cosine loss yields better AD and anomaly

Fig. 8. Illustration of the different ways to transfer distilled knowledge to T_i .

segmentation performance than MSE loss. Interestingly, for “textured” images, using MSE loss proves to be more effective. In this article, ADPS uses the experimental setup of the second row, that is, it uses the cosine similarity loss and computes $\mathcal{W}_i$ by cosine similarity.

E. Ablation Study on the Framework of ADPS

To investigate each component of the proposed ADPS framework, we compare ADPS with its variants on the MVtec-AD dataset.

- 1) *W/O. PW*: The proposed ADPS without PSM and WMB.
- 2) *W/O. T*: The proposed ADPS without the teacher network, that is, direct training the student model and PSM.
- 3) *W/O. S*: The proposed ADPS removes the student network and directly uses a pretrained teacher network. In this setting, only the parameters of the PSB can be learned.

We show the variant models and their results in Table VIII. Based on the experimental results, the following conclusions can be made.

1) *Pretrained Knowledge is Essential for the Teacher Network*: The comparison results from *W/O. T* and *W/O. S* show a significant drop in AD and anomaly segmentation performance for multiple categories. Especially, the anomaly segmentation performance decreases by 10% for AP_{seg} . The experimental results confirm the effectiveness of prior knowledge from the pretrained model for AD and emphasize the need for further exploration. Using only the teacher network is intuitively more effective than using only the student network, as the pretrained model provides a powerful feature representation for the PSB. Another possible explanation is that after fixing the image encoder (teacher model), the PSM module is more capable of being trained more efficiently [74], yielding more accurate segmentation results.

TABLE VI
ANOMALY DETECTION RESULTS OF DIFFERENT T-S FEATURE FUSION METHODS ON THE MVTEC AD DATASET.
WMB ACHIEVES THE BEST PERFORMANCE. “_” INDICATES SUBOPTIMAL RESULTS

	$AUROC_{cla}$			$AUROC_{seg}$			PRO_{seg}			AP_{seg}		
	Texture	Object	Mean	Texture	Object	Mean	Texture	Object	Mean	Texture	Object	Mean
Method(1)	99.8	<u>99.2</u>	<u>82.3</u>	<u>94.6</u>	<u>95.7</u>	<u>69.1</u>	97.8	<u>88.9</u>	<u>91.9</u>	<u>96.4</u>	<u>96.9</u>	<u>73.5</u>
Method(2)	99.6	99.0	81.7	94.5	92.8	66.8	97.7	85.0	89.2	96.1	94.9	71.8
WMB	<u>98.2</u> ↓	99.5 ↑	83.1 ↑	97.4 ↑	97.0 ↑	74.3 ↑	97.8 ↑	90.6 ↑	93.0 ↑	97.7 ↑	97.8 ↑	77.2 ↑

TABLE VII
ABLATION STUDY ON THE SELECTION OF DIFFERENT KNOWLEDGE DISTILLATION LOSS FUNCTIONS AND DIFFERENT GENERATION METHODS OF $\mathcal{W}_i$

Loss		$\mathcal{W}_i$		$AUROC_{cla}$			$AUROC_{seg}$			AP_{seg}		
MSE	Cosine	$\ T_i - S_i\ _2$	$Cosine(T_i, S_i)$	Texture	Object	Mean	Texture	Object	Mean	Texture	Object	Mean
✓	×	✓	×	99.4	91.5	94.1	99.8	92.4	94.5	77.3	56.6	63.5
×	✓	×	✓	98.3	97.4	97.7	99.5	97.0	97.8	83.1	74.3	77.2
✓	✓	✓	×	99.2	95.9	97.0	99.4	95.6	96.8	83.4	71.0	75.2
✓	✓	×	✓	99.2	96.6	97.5	99.5	96.0	97.2	83.3	71.8	75.6

TABLE VIII
PERFORMANCE OF FOUR STRUCTURES ON THE MVTEC AD DATASET. ADPS IS THE BEST IN TERMS OF
ANOMALY CLASSIFICATION AND ANOMALY SEGMENTATION PERFORMANCE

Category	AP_{seg}				$AUROC_{cla}$				$AUROC_{seg}$			
	W/O.PW	W/O.T	W/O.S	ADPS	W/O.PW	W/O.T	W/O.S	ADPS	W/O.PW	W/O.T	W/O.S	ADPS
Capsule	28.9	27.4	56.1	62.6	70.5	88.2	97.2	96.3	97.7	88.0	93.0	97.2
Bottle	82.8	60.0	89.7	93.9	100.0	91.8	100.0	100.0	98.7	93.6	99.0	99.5
Carpet	66.4	73.6	84.3	83.7	95.7	100.0	99.6	97.0	99.0	98.8	99.6	99.5
Leather	68.9	81.4	80.6	80.4	99.8	100.0	100.0	100.0	99.7	99.9	99.9	99.8
Pill	65.2	78.3	89.6	85.9	77.6	89.8	94.8	94.7	96.7	98.3	99.3	99.0
Transistor	29.3	30.1	28.9	48.3	85.4	82.5	89.8	95.5	80.6	74.2	66.3	84.8
Tile	82.1	95.2	96.5	96.1	98.6	99.6	100.0	99.7	97.5	99.5	99.6	99.5
Cable	49.7	43.5	51.3	63.3	86.6	78.9	91.2	94.9	95.6	86.2	89.7	94.1
Zipper	57.7	66.4	85.6	86.0	94.5	99.9	100.0	100.0	98.2	98.8	99.6	99.5
Toothbrush	40.6	47.2	67.0	65.7	71.1	86.1	98.1	99.4	98.0	96.7	99.2	99.2
Metal_nut	82.1	90.4	58.3	87.1	99.6	99.2	99.6	100.0	98.2	98.4	89.3	98.1
Hazelnut	77.9	78.8	86.5	90.5	99.9	93.6	99.7	99.1	99.2	99.1	99.5	99.6
Screw	22.5	21.8	52.8	59.2	84.9	79.5	91.6	94.2	98.5	96.5	97.4	98.8
Grid	36.7	56.6	62.9	65.6	94.6	99.2	99.8	100.0	97.9	99.3	99.3	99.5
Wood	77.9	78.4	88.4	89.8	99.0	97.9	98.0	94.6	97.9	96.8	98.5	99.0
MEAN	57.9	61.9	71.9	77.2	90.5	92.4	97.3	97.7	96.9	94.9	95.3	97.8

2) *Exploitation Potential of the Teacher Network Is Significant*: Comparing with ADPS, W/O. PW shows a decrease of 7% in $AUROC_{cla}$ and 20% in AP_{seg} , highlighting the effectiveness and potential of the PSM and WMB modules in utilizing the features extracted by the teacher network. While the teacher network only serves as an extractor of reference features without leveraging its powerful feature representation capability, introducing WMB to utilize distilled knowledge and PSM to fully exploit the features extracted by the teacher model enhances the AD capability. Table VI also demonstrates the significant impact of different methods that employ pretrained knowledge and distilled knowledge on AD. The proposed WMB may not be the optimal scheme, and future research could explore methods that combine teacher networks and distillation strategies to further enhance AD ability.

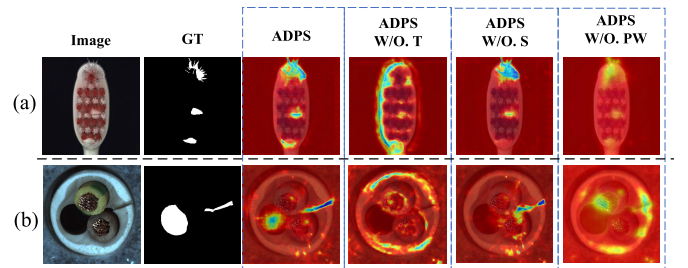


Fig. 9. Anomaly heatmaps for four structures. ADPS captures more tiny anomaly regions with clearer and more detailed decision boundaries. (a) and (b) represent two cases of ablation studies in ADPS.

3) *Asymmetric Distillation and WMB Contribute to the Performance of ADPS*: Comparing W/O. S with ADPS, it can be observed that distilled knowledge obtained through

asymmetric distillation significantly improves the performance of ADPS. However, without the WMB, the distilled knowledge cannot be effectively transferred to the teacher network. The corresponding results demonstrate that fully integrating pre-trained prior knowledge and distilled knowledge is effective in improving the performance of AD. The insight behind this is that prior knowledge provides powerful feature representation, while distilled knowledge enhances the discrimination between anomalies and normal instances.

a) Feature visualization: We present visualizations in Fig. 9, which includes two examples that compare anomaly segmentation results. Fig. 9(a) and (b) illustrates the boundary between normal and abnormal features learned by *W/O. S* is not accurate. Additionally, *W/O. PW* shows the inability to derive the decision boundary between abnormal and normal from low-resolution feature maps, which leads to inefficient detection results. The limitations of *W/O. S* in capturing differences in small anomalous regions are attributed to the lack of an asymmetrical distillation model that focuses on local regions. On the other hand, Fig. 9 demonstrates that $\mathcal{AP}_{\text{seg}}$ is more indicative of the performance of anomalous segmentation (e.g., although the segmentation results of *W/O. PS* have many errors, $\text{AUROC}_{\text{seg}}$ still maintains a high performance).

b) Discussion: Undoubtedly, the PSM plays a vital role, particularly in the anomaly segmentation capability of ADPS. It further confirms our hypothesis that solely relying on acquiring high-precision segmentation maps from low-resolution feature map differences is extremely challenging. Consequently, solving the anomaly segmentation challenge may have to rely on high-resolution images. Additionally, the approach employed by ADPS to exploit the features extracted by the pretrained model is straightforward. We believe that more effective distilled knowledge can be obtained by leveraging the discrepancy between the learning capabilities of the teacher network and the student network.

V. CONCLUSION

In this article, we presented a novel knowledge distillation-based framework, ADPS, to alleviate the main problems of existing KDAD methods for image AD. Our proposed ADPS framework establishes an asymmetric distillation paradigm with asymmetric data streams to mitigate the problem of the student network over-approximating the feature output of the teacher network when anomalous images are input. The asymmetric distillation paradigm further utilizes local patches as inputs to facilitate the student model's ability to capture local details. Additionally, customized WMB and PSM were developed to fully incorporate the distilled posterior knowledge and pretrained prior knowledge to produce high-quality segmentation maps with fine structures and clear boundaries. Extensive experiments on three benchmarks demonstrate that our ADPS framework performs favorably against state-of-the-art methods. Importantly, our simple framework is modular, promising to serve as a strong baseline for future efficient KDAD research.

Despite these advancements, ADPS's reliance on simulated anomalous samples remains a limitation. Moving forward, our research will focus on reducing this dependency and exploring

the application of ADPS in additional industrial contexts, potentially broadening its scope and impact.

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Jinhui Tang (Senior Member, IEEE) received the B.E. and Ph.D. degrees from the University of Science and Technology of China, Hefei, China, in 2003 and 2008, respectively.

He is currently a Professor with Nanjing University of Science and Technology, Nanjing, China. He has authored or coauthored more than 200 articles in top-tier journals and conferences. His research interests include multimedia analysis and computer vision.

Dr. Tang was a recipient of the Best Paper Awards from ACM MM 2007 and ACM MM Asia 2020, the Best Paper Runner-Up in ACM MM 2015. He has served as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT). He is a Fellow of IAPR.



Peng Xing is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

His current research interests include anomaly detection and unsupervised deep learning.



Hao Tang received the B.E. degree from Harbin Engineering University, Harbin, China, in 2018. He is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

His current research interests include fine-grained image analysis, data-efficient learning, and multi-modal learning.



Zechao Li (Senior Member, IEEE) received the B.E. degree from the University of Science and Technology of China, Hefei, China, in 2008, and the Ph.D. degree from the National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2013.

He is currently a Professor with Nanjing University of Science and Technology, Nanjing, China. His research interests include big media analysis, computer vision, and so on.

Dr. Li was a recipient of the best paper award in ACM Multimedia Asia 2020, and the best student paper award in ICIMCS 2018. He serves as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS).